Time Series Practice Questions

1. You are given the following time-series for monthly building settlement (mm):
   5, 7, 10, 12, 15, 14, 12, 9, 7, 6, 7, 9
   Identify whether each of these components exist in this time series: trend, seasonality and noise. Approximately sketch the decomposition.
   Answer: Trend: weak
   Seasonality: Yes (rising till 5th month) and then falls
   Noise: Yes

2. You are working on modelling displacement of bridge girder. You have collected the following data:

$$t: 0,1,2,3,4$$
$$x(t): 10,13,15,18,20$$

   Assuming linear trend model $x(t) = a + bt + \epsilon_t$, find the Least-Square Estimate for a and b. What type of error are we implicitly assuming here?
   Answer: Design matrix, $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}, x = \begin{bmatrix} a \\ b \end{bmatrix}$ and $y = \begin{bmatrix} 10 \\ 13 \\ 15 \\ 18 \\ 20 \end{bmatrix}$

$$\hat{a} = 10.2 \; and \; \hat{b} = 2.5$$

   Since we are using LS as best estimator, we are assuming that the noise is Gaussian White Noise.

3. Given the following observations:

$$y = Ax + \epsilon$$

$$y = \begin{bmatrix} 5 \\ 8 \\ 6 \end{bmatrix}, A = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, cov(\epsilon) = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 4 \end{bmatrix}$$

   Compute the BLUE of x.
   Answer:

$$\hat{x}_{BLUE} = (A^T \Sigma^{-1} A)^{-1} A^T \Sigma^{-1} y = 6.13$$

4. In question 3, compute the LS estimate. How do the two compare?
   Answer:

$$\hat{x}_{LS} = (A^T A)^{-1} A^T y = 6.33$$